DIB isolators, EEPROM, ADC, MCU

File: DIB_EEPROM_ADC_MCU.kicad_sch

DAC, OE, Vref, IOEXP, Temp

File: DAC_Vref_IOEXP_Temp.kicad_sch

Power pre-regulator

File: Power_pre-regulator.kicad_sch

Bias power

File: Bias_power.kicad_sch

Post-regulator

File: Post-regulator.kicad_sch

Rsense protection, power output, Rprog

File: Rsense_protection_power_output.kicad_sch

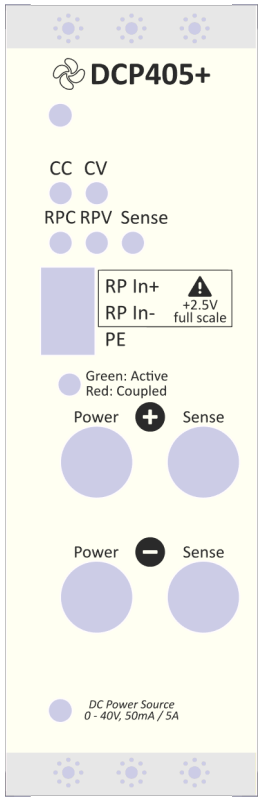
Mode detection, OVP

File: CC_CV_OVP.kicad_sch

Front mask details:

Material: Aluminum PCB
Thickness: 1.6 mm
PCB layout: 1-layer

Project repository: https://github.com/eez-open/dib-dcp405plus/blob/main/KiCad/EEZ%20DIB%20DCP405plus%20front%20mask.kicad_pro



KK1
M3X8/D7985B

KK2
M3X8/D7985B

KK3
M3X8/D7985B

KK4
M3X8/D7985B

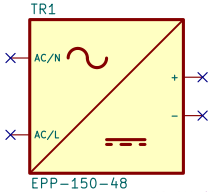
KK5
SK96/84SA

KK6
SP400-0.007-00-104

KK8
B3/BN2842

KK7
M3X8/D7985B

TR1



EPP-150-48
Alternative: Mean Well LOP-200-48

KK9
PCB-MB-01

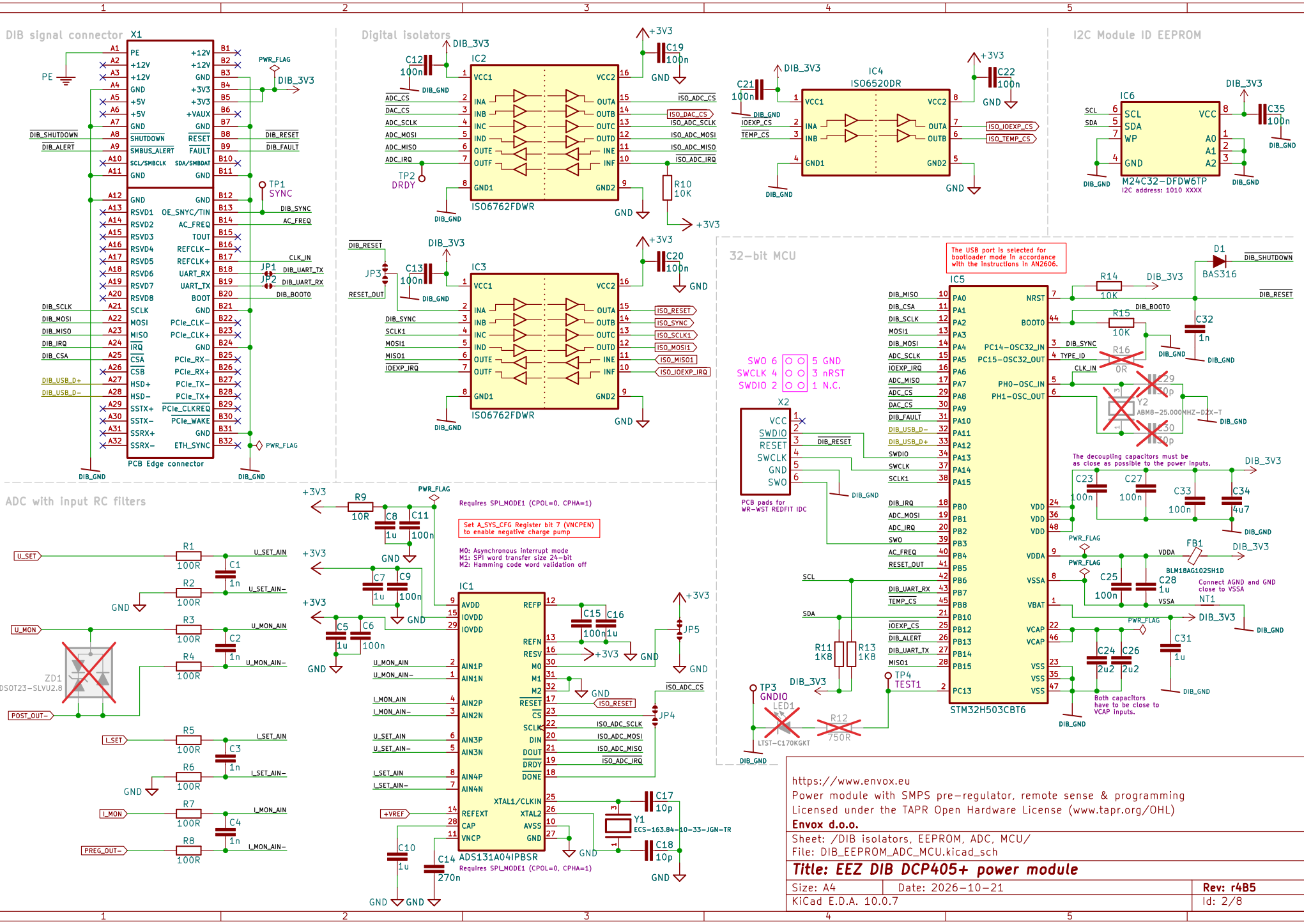
KK10
PCB-MB-01

https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

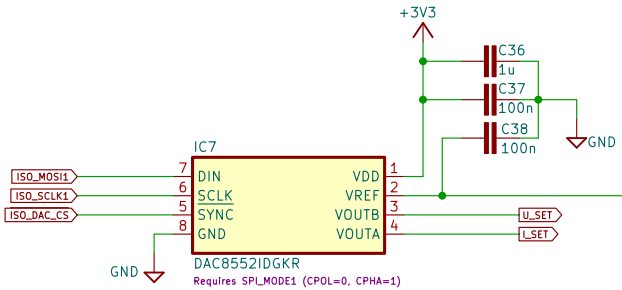
Sheet: /
File: EEZ DIB DCP405plus.kicad_sch

Title: EEZ DIB DCP405+ power module

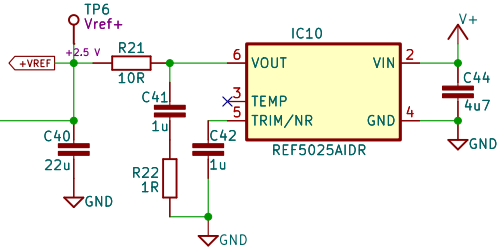
Size: A4	Date: 2026-10-21	**Rev: r4B5**
KiCad E.D.A. 10.0.7		Id: 1/8


DIB signal connector
X1
PCB Edge connector
Digital isolators
IC2
ISO6762FDWR
IC3
ISO6762FDWR
IC4
ISO6520DR
I2C Module ID EEPROM
IC6
M24C32-DFDW6TP
I2C address: 1010 XXXX
32-bit MCU
IC5
STM32H503CBT6
The USB port is selected for bootloader mode in accordance with the instructions in AN2606.
PCB pads for WR-WST REDFIT IDC
The decoupling capacitors must be as close as possible to the power inputs.
Both capacitors have to be close to VCAP inputs.
Connect AGND and GND close to VSSA
ADC with input RC filters
IC1
ADS131A04IPBSR
Requires SPI_MODE1 (CPOL=0, CPHA=1)
Set A_SYS_CFG Register bit 7 (VNCPEN) to enable negative charge pump
M0: Asynchronous interrupt mode
M1: SPI word transfer size 24-bit
M2: Hamming code word validation off
https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /DIB isolators, EEPROM, ADC, MCU/
File: DIB_EEPROM_ADC_MCU.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-21
Rev: r4B5
KiCad E.D.A. 10.0.7
Id: 2/8

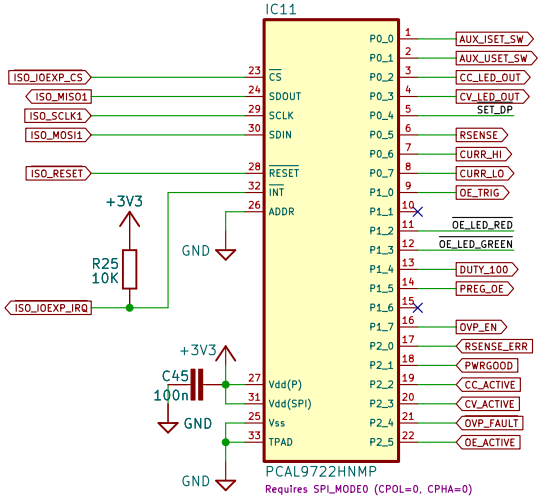
16-bit DAC (set voltage and current)



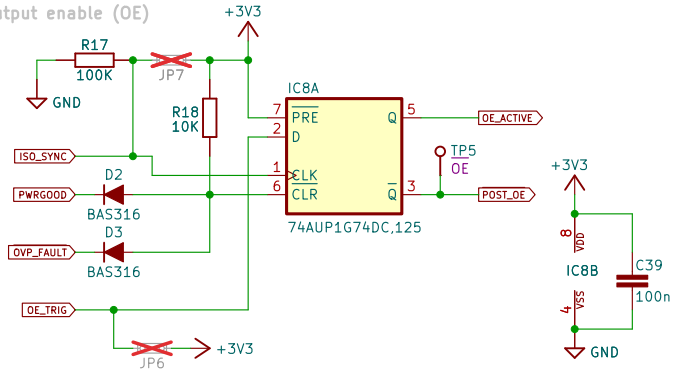
Voltage reference



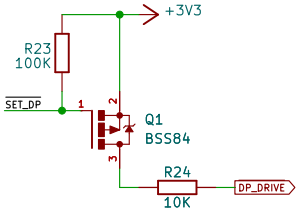
SPI 22-bit I/O Expander



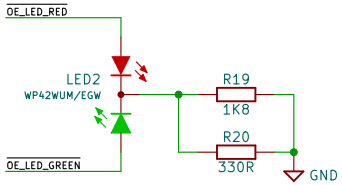
Output enable (OE)



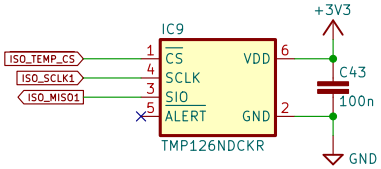
Down-programmer



OE indicator



SPI Temperature sensor

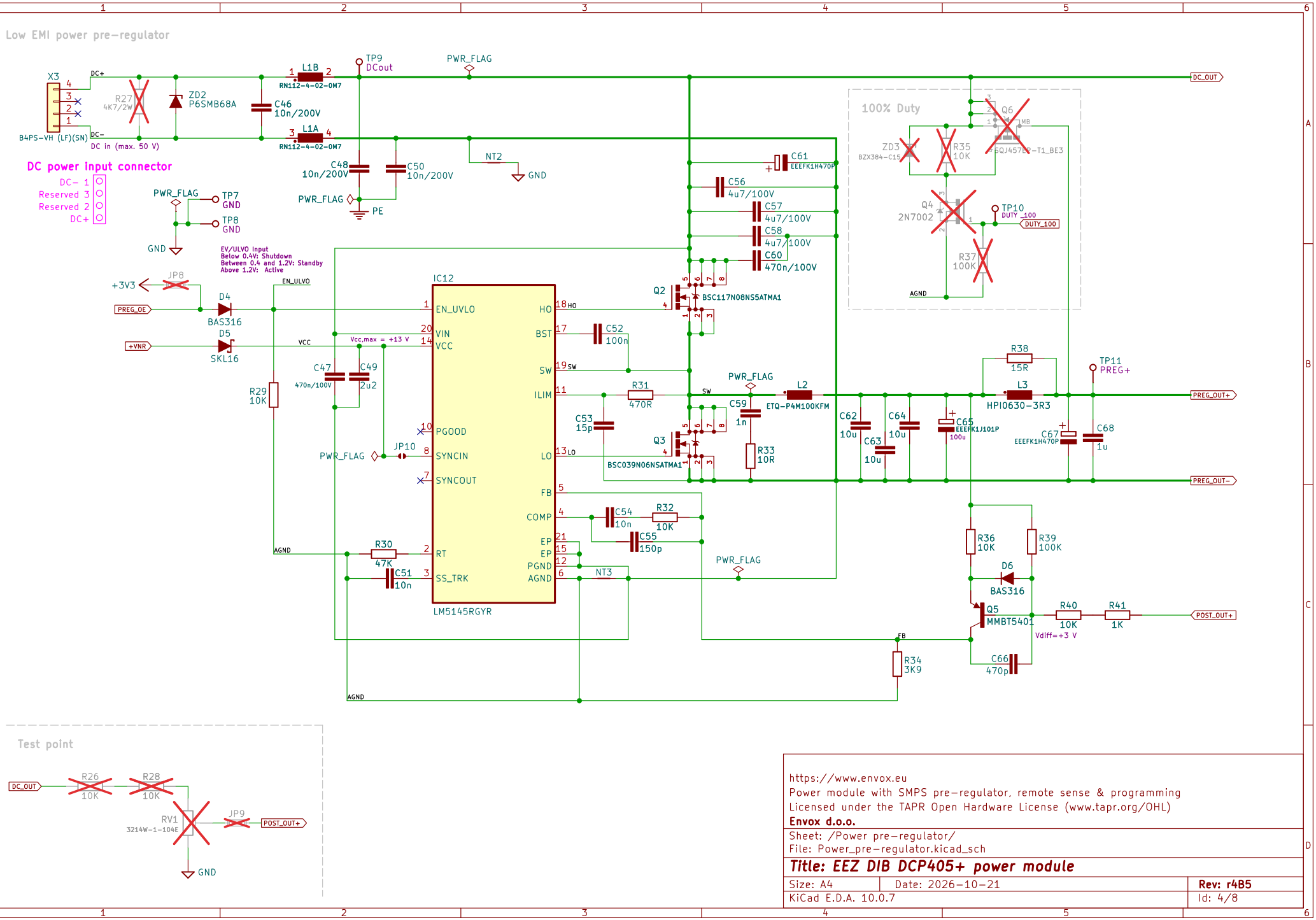


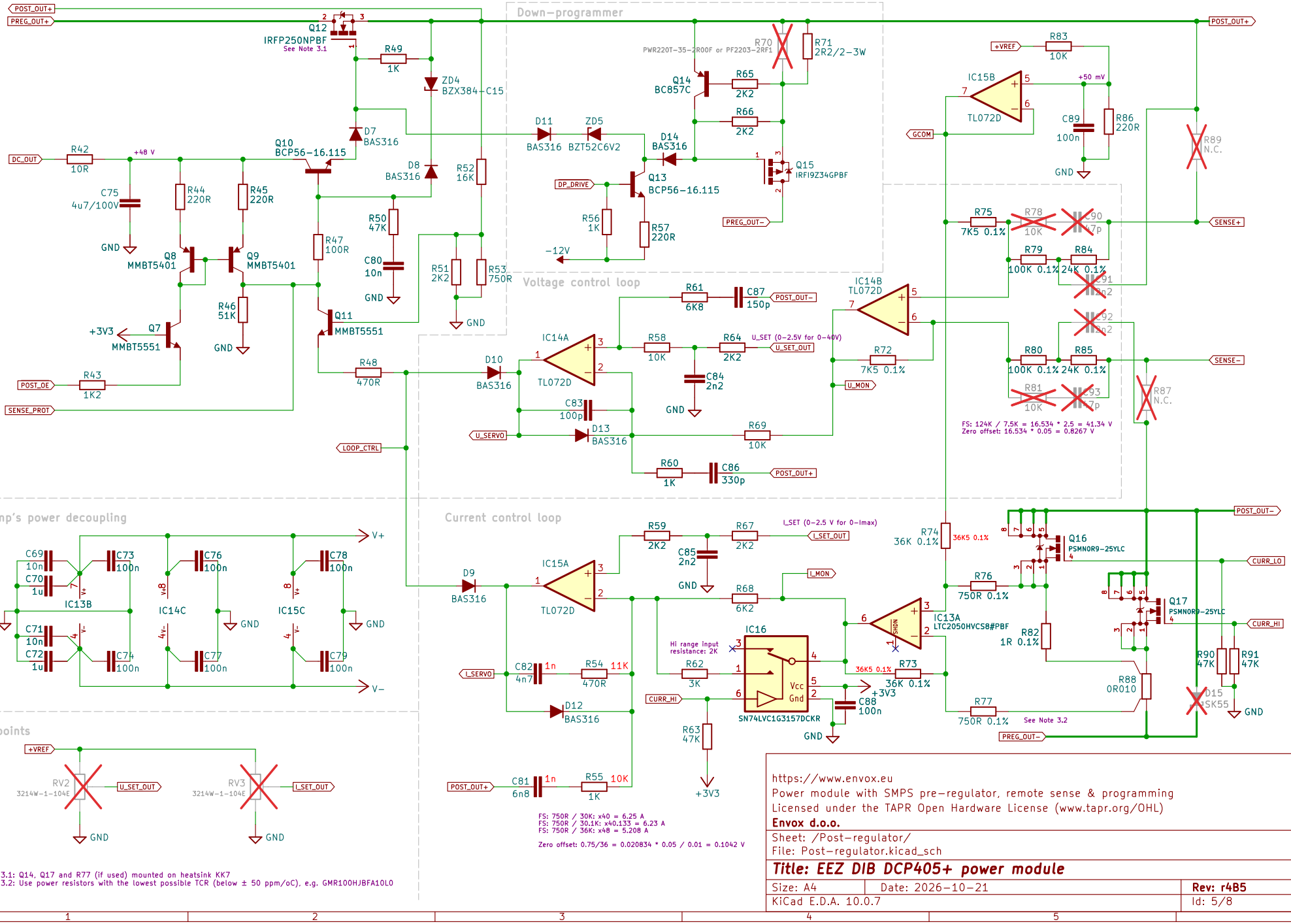
https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

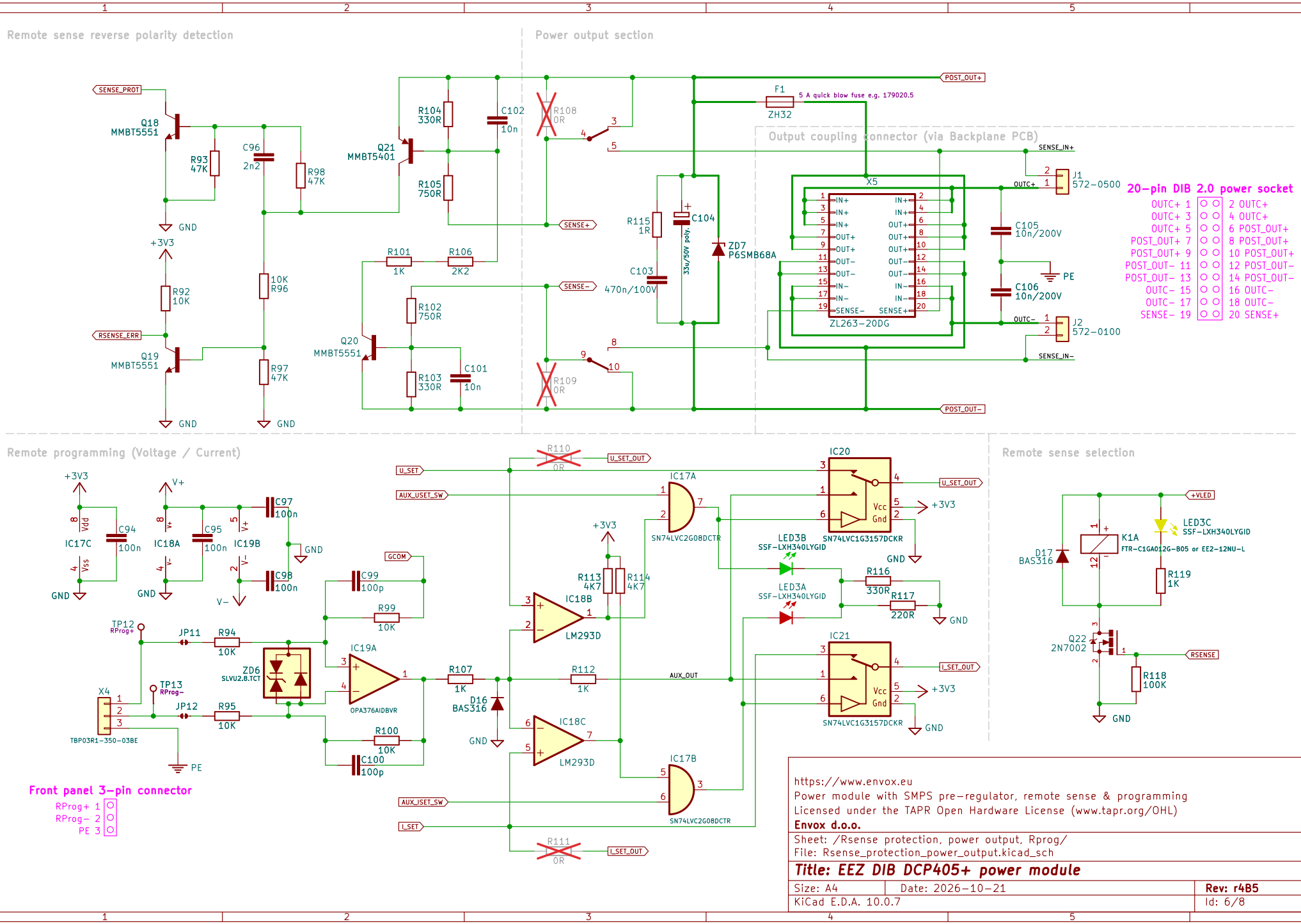
Sheet: /DAC, OE, Vref, IOEXP, Temp/
File: DAC_Vref_IOEXP_Temp.kicad_sch

Title: EEZ DIB DCP405+ power module

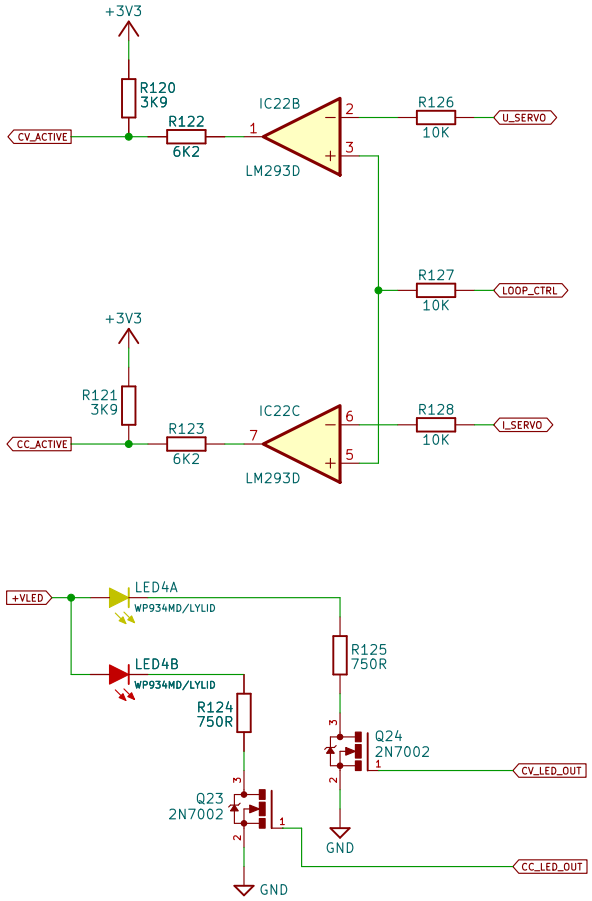
Size: A4	Date: 2026-10-21	**Rev: r4B5**
KiCad E.D.A. 10.0.7		Id: 3/8


Low EMI power pre-regulator
DC power input connector
DC- 1
Reserved 3
Reserved 2
DC+
X3
B4PS-VH (LF)(SN)
DC in (max. 50 V)
R27 4K7/2W
ZD2 P6SMB68A
C46 10n/200V
L1B RN112-4-02-0M7
L1A RN112-4-02-0M7
TP9 DCout
PWR_FLAG
C48 10n/200V
C50 10n/200V
PE
NT2
GND
TP7 GND
TP8 GND
JP8
+3V3
PREG_OE
D4 BAS316
+VNR
D5 SKL16
EN_UVLO
VCC
Vcc,max = +13 V
EV/ULVO Input
Below 0.4V: Shutdown
Between 0.4 and 1.2V: Standby
Above 1.2V: Active
C47 470n/100V
C49 2u2
R29 10K
IC12
EN_UVLO
VIN
VCC
PGOOD
SYNCIN
SYNCOUT
RT
SS_TRK
HO
BST
SW
ILIM
LO
FB
COMP
EP
EP
PGND
AGND
LM5145RGYR
JP10
R30 47K
C51 10n
C52 100n
R31 470R
C53 15p
C54 10n
R32 10K
C55 150p
NT3
AGND
Q2 BSC117N08NS5ATMA1
Q3 BSC039N06NSATMA1
C61 EEEFK1H470P
C56 4u7/100V
C57 4u7/100V
C58 4u7/100V
C60 470n/100V
C59 1n
R33 10R
L2 ETQ-P4M100KFM
C62 10u
C63 10u
C64 10u
C65 EEEFK1J101P 100u
C67 EEEFK1H470P
C68 1u
R38 15R
L3 HPI0630-3R3
TP11 PREG+
DC_OUT
PREG_OUT+
PREG_OUT-
100% Duty
ZD3 BZX384-C15
R35 10K
Q6 SQJ457EP-T1_BE3
Q4 2N7002
TP10 DUTY_100
R37 100K
R36 10K
R39 100K
D6 BAS316
Q5 MMBT5401
R40 10K
R41 1K
POST_OUT+
Vdiff=+3 V
R34 3K9
C66 470p
FB
Test point
R26 10K
R28 10K
RV1 3214W-1-104E
JP9
https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /Power pre-regulator/
File: Power_pre-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-21
Rev: r4B5
KiCad E.D.A. 10.0.7
Id: 4/8

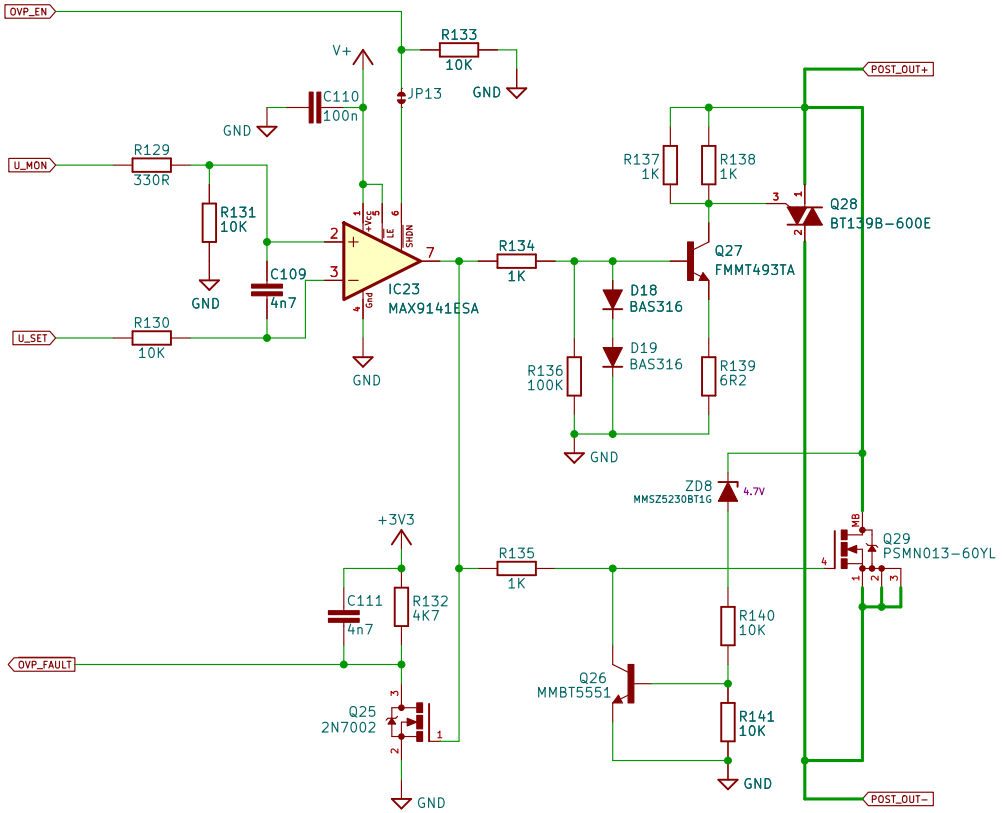

Down-programmer
Voltage control loop
Current control loop
Op amp's power decoupling
Test points
POST_OUT+
PREG_OUT+
DC_OUT
+48 V
Q12
IRFP250NPBF
See Note 3.1
Q10
BCP56-16.115
Q8
MMBT5401
Q9
MMBT5401
Q7
MMBT5551
Q11
MMBT5551
Q13
BCP56-16.115
Q14
BC857C
Q15
IRFI9Z34GPBF
Q16
PSMN0R9-25YLC
Q17
PSMN0R9-25YLC
D7
BAS316
D8
BAS316
ZD4
BZX384-C15
D11
BAS316
ZD5
BZT52C6V2
D14
BAS316
D10
BAS316
D9
BAS316
D12
BAS316
D13
BAS316
D15
PSK55
R70
PWR220T-35-2R00F or PF2203-2RF1
R71
2R2/2-3W
R42
10R
R43
1K2
R44
220R
R45
220R
R46
51K
R47
100R
R48
470R
R49
1K
R50
47K
R51
2K2
R52
16K
R53
750R
R54
470R
11K
R55
10K
1K
R56
1K
R57
220R
R58
10K
R59
2K2
R60
1K
R61
6K8
R62
3K
R63
47K
R64
2K2
R65
2K2
R66
2K2
R67
2K2
R68
6K2
R69
10K
R72
7K5 0.1%
R73
36K 0.1%
R74
36K 0.1%
36K5 0.1%
R75
7K5 0.1%
R76
750R 0.1%
R77
750R 0.1%
R78
10K
R79
100K 0.1%
R80
100K 0.1%
R81
10K
R82
1R 0.1%
R83
10K
R84
24K 0.1%
R85
24K 0.1%
R86
220R
R87
N.C.
R88
0R010
R89
N.C.
R90
47K
R91
47K
C69
10n
C70
1u
C71
10n
C72
1u
C73
100n
C74
100n
C75
4u7/100V
C76
100n
C77
100n
C78
100n
C79
100n
C80
10n
C81
6n8
1n
C82
4n7
1n
C83
100p
C84
2n2
C85
2n2
C86
330p
C87
150p
C88
100n
C89
100n
C90
47p
C91
2n2
C92
2n2
C93
47p
IC13B
IC14C
IC15C
IC14A
TL072D
IC15B
TL072D
IC14B
TL072D
IC13A
LTC2050HVCS8#PBF
IC16
SN74LVC1G3157DCKR
RV2
3214W-1-104E
RV3
3214W-1-104E
+VREF
+50 mV
+3V3
-12V
GND
GCOM
SENSE+
SENSE-
SENSE_PROT
POST_OE
LOOP_CTRL
U_SET_OUT
U_SET (0-2.5V for 0-40V)
U_MON
I_SET_OUT
I_SET (0-2.5 V for 0-Imax)
I_MON
I_SERVO
U_SERVO
DP_DRIVE
PREG_OUT-
POST_OUT-
CURR_LO
CURR_HI
Hi range input resistance: 2K
FS: 124K / 7.5K = 16.534 * 2.5 = 41.34 V
Zero offset: 16.534 * 0.05 = 0.8267 V
FS: 750R / 30K: x40 = 6.25 A
FS: 750R / 30.1K: x40.133 = 6.23 A
FS: 750R / 36K: x48 = 5.208 A
Zero offset: 0.75/36 = 0.020834 * 0.05 / 0.01 = 0.1042 V
See Note 3.2
Note 3.1: Q14, Q17 and R77 (if used) mounted on heatsink KK7
Note 3.2: Use power resistors with the lowest possible TCR (below ± 50 ppm/oC), e.g. GMR100HJBFA10L0
https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /Post-regulator/
File: Post-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-21
Rev: r4B5
KiCad E.D.A. 10.0.7
Id: 5/8


Remote sense reverse polarity detection
Power output section
Output coupling connector (via Backplane PCB)
Remote programming (Voltage / Current)
Remote sense selection
Front panel 3-pin connector
RProg+ 1
RProg- 2
PE 3
20-pin DIB 2.0 power socket
OUTC+ 1
2 OUTC+
OUTC+ 3
4 OUTC+
OUTC+ 5
6 POST_OUT+
POST_OUT+ 7
8 POST_OUT+
POST_OUT+ 9
10 POST_OUT+
POST_OUT- 11
12 POST_OUT-
POST_OUT- 13
14 POST_OUT-
OUTC- 15
16 OUTC-
OUTC- 17
18 OUTC-
SENSE- 19
20 SENSE+
https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /Rsense protection, power output, Rprog/
File: Rsense_protection_power_output.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-21
Rev: r4B5
KiCad E.D.A. 10.0.7
Id: 6/8

Constant Voltage (CV) and Constant Current (CC) mode detection and indication



Over-voltage protection (OVP)



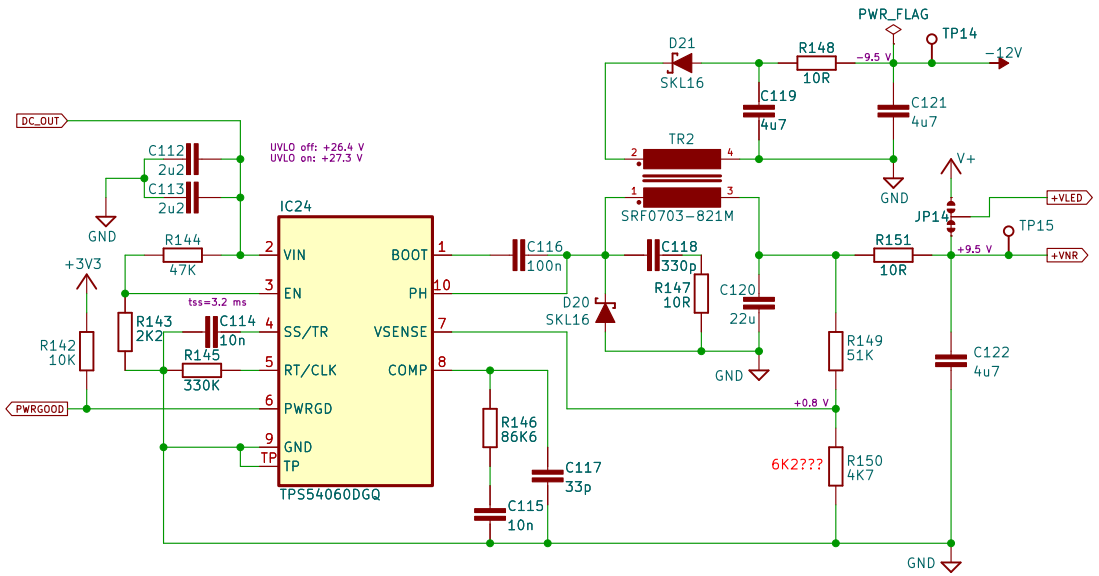
https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Mode detection, OVP/
File: CC_CV_OVP.kicad_sch

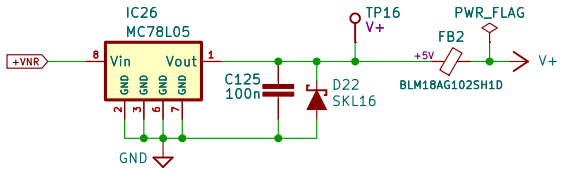
Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-10-21	**Rev: r4B5**
KiCad E.D.A. 10.0.7		Id: 7/8

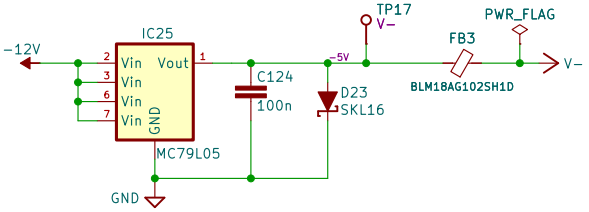
Bias power supply



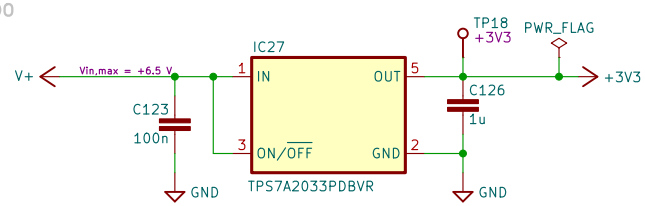
+5V LDO



-5V LDO



+3V3 LDO



https://www.envox.eu
Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Bias power/
File: Bias_power.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-10-21	**Rev: r4B5**
KiCad E.D.A. 10.0.7		Id: 8/8